TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	53.4 ± 0.66
Correlation Stability	64.6 ± 2.20
Distribution Consistency	50.0 ± 0.00
Entropy Preservation	10.0 ± 0.00
Mean Z Score Drift	2.1 ± 0.15
Categorical Fidelity	94.8 ± 5.56
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	100.0 ± 0.00
Feature Accuracy	-29.3 ± 9.53
Novelty Score	1.0 ± 0.01

TabuLLM: AI Benchmark Report

Dataset Audit: breast_cancer_wisconsin_diagnostic.csv

1. Dataset Portrait

Attribute	Value
Rows	569
Columns	31
Numeric features	30
Categorical features	1
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
radius1	14.127	10.510	-3.617	25.6%	Drift
texture1	19.290	22.578	+3.288	17.0%	Warn
perimeter1	91.969	103.582	+11.613	12.6%	Warn
area1	654.889	1283.390	+628.501	96.0%	Drift
smoothness1	0.096	0.120	+0.024	24.5%	Warn
compactness1	0.104	0.256	+0.151	144.9%	Drift
concavity1	0.089	0.153	+0.064	72.2%	Drift
concave_points1	0.049	0.120	+0.071	144.9%	Drift
symmetry1	0.181	0.227	+0.046	25.3%	Drift
fractal_dimension1	0.063	0.090	+0.027	42.7%	Drift
radius2	0.405	1.289	+0.884	218.1%	Drift
texture2	1.217	3.473	+2.256	185.4%	Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	53.4 ± 0.66
Correlation Stability (*2)	64.6 ± 2.20
Distribution Consistency (*3)	50.0 ± 0.00
Entropy Preservation (*4)	10.0 ± 0.00
Mean Z Score Drift (*5)	2.1 ± 0.15
Categorical Fidelity (*6)	94.8 ± 5.56
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	100.0 ± 0.00
Feature Accuracy (*10)	-29.3 ± 9.53
Novelty Score (*11)	1.0 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9429824561403507	0.6692982456140351	0.9429824561403507	0	
F1	0.934337828992506	0.13144912803280792	0.934337828992506	0	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9692982456140351	0.6263157894736843	0.9675438596491228	-0.001754385964912272	
F1	0.9659514824350456	0.05986394557823129	0.9649605942390138	-0.000990888196031780 5	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9394736842105263	0.7061403508771928	0.7877192982456139	-0.15175438596491225	*
F1	0.9314283604116784	0.2709228296495315 7	0.7105369813486824	-0.2208913790629962	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9368421052631586	0.6543859649122806	0.9271929824561402	-0.009649122807017551	
F1	0.9282287981129432	0.2775000204964233 6	0.9200970720636763	-0.008131726049266808	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9333333333333333	0.6736842105263158	0.9184210526315789	-0.014912280701754387	*
F1	0.9242527459470589	0.1416347220944112	0.9070023248177739	-0.017250421129285133	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9710526315789474	0.6921052631578948	0.9526315789473683	-0.01842105263157897	*
F1	0.9676338620041433	0.2391379465031185	0.9460768879657188	-0.021556974038424593	*

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9578947368421051	0.6824561403508771	0.9596491228070174	0.001754385964912262	
F1	0.9521479233175448	0.2048000480288121	0.9548988424633672	0.002750919145822194	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9605263157894737	0.6491228070175439	0.9622807017543864	0.001754385964912264	
F1	0.9557853896138416	0.2289168802774833	0.9578939159138454	0.002108526300003294	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9508771929824562	0.6491228070175439	0.7973684210526317	-0.15350877192982454	*
F1	0.9457538157039082	0.2438353591177486	0.6932235486688298	-0.25253026703507836	*

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9263157894736842	0.6333333333333334	0.8052631578947368	-0.12105263157894738	*
F1	0.9142153956337035	0.3393464842000460	0.7714794447189167	-0.14273595091478683	*

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	17.99	10.38	122.8	1001.0	0.1184	0.2776	0.3001	0.1471	0.2419	0.07871
2	20.57	17.77	132.9	1326.0	0.08474	0.07864	0.0869	0.07017	0.1812	0.05667
3	19.69	21.25	130.0	1203.0	0.1096	0.1599	0.1974	0.1279	0.2069	0.05999
4	11.42	20.38	77.58	386.1	0.1425	0.2839	0.2414	0.1052	0.2597	0.09744
5	20.29	14.34	135.1	1297.0	0.1003	0.1328	0.198	0.1043	0.1809	0.05883
6	12.45	15.7	82.57	477.1	0.1278	0.17	0.1578	0.08089	0.2087	0.07613
7	18.25	19.98	119.6	1040.0	0.09463	0.109	0.1127	0.074	0.1794	0.05742
8	13.71	20.83	90.2	577.9	0.1189	0.1645	0.09366	0.05985	0.2196	0.07451
9	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
10	12.46	24.04	83.97	475.9	0.1186	0.2396	0.2273	0.08543	0.203	0.08243
11	16.02	23.24	102.7	797.8	0.08206	0.06669	0.03299	0.03323	0.1528	0.05697
12	15.78	17.89	103.6	781.0	0.0971	0.1292	0.09954	0.06606	0.1842	0.06082
13	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
14	15.85	23.95	103.7	782.7	0.08401	0.1002	0.09938	0.05364	0.1847	0.05338
15	13.73	22.61	93.6	578.3	0.1131	0.2293	0.2128	0.08025	0.2069	0.07682
16	14.54	27.54	96.73	658.8	0.1139	0.1595	0.1639	0.07364	0.2303	0.07077
17	14.68	20.13	94.74	684.5	0.09867	0.072	0.07395	0.05259	0.1586	0.05922
18	16.13	20.68	108.1	798.8	0.117	0.2022	0.1722	0.1028	0.2164	0.07356
19	19.81	22.15	130.0	1260.0	0.09831	0.1027	0.1479	0.09498	0.1582	0.05395
20	13.54	14.36	87.46	566.3	0.09779	0.08129	0.06664	0.04781	0.1885	0.05766

B.1 Audit Trace: Fold 1 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	39.28	43.79	143.5	0.1634	0.3454	0.0	0.0	0.304	0.09744
2	6.981	9.71	43.79	143.5	0.1634	0.3454	0.4268	0.2012	0.1167	0.09744
3	6.981	39.28	43.79	143.5	0.05263	0.02344	0.0	0.0	0.304	0.09744
4	6.981	9.71	43.79	143.5	0.1634	0.3454	0.4268	0.2012	0.1167	0.09744
5	28.11	9.71	188.5	143.5	0.1634	0.02344	0.0	0.2012	0.1167	0.09744
6	6.981	9.71	188.5	2501.0	0.1634	0.3454	0.4268	0.2012	0.304	0.09744
7	13.24	20.13	86.87	542.9	0.08284	0.1223	0.101	0.02833	0.1601	0.06432
8	11.2	29.37	70.67	386.0	0.07449	0.03558	0.0	0.0	0.106	0.05502
9	16.74	21.59	110.1	869.5	0.0961	0.1336	0.1348	0.06018	0.1896	0.05656
10	13.17	18.66	85.98	534.6	0.1158	0.1231	0.1226	0.0734	0.2128	0.06777

B.2 Audit Trace: Fold 2 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	39.28	43.79	143.5	0.05263	0.01938	0.0	0.0	0.304	0.09744
2	6.981	9.71	43.79	143.5	0.1425	0.3114	0.0	0.0	0.106	0.04996
3	27.42	9.71	43.79	143.5	0.05263	0.3114	0.0	0.1913	0.106	0.09744
4	6.981	9.71	43.79	143.5	0.05263	0.3114	0.0	0.0	0.106	0.04996
5	6.981	9.71	43.79	2501.0	0.05263	0.01938	0.4264	0.0	0.106	0.04996
6	27.42	39.28	43.79	143.5	0.1425	0.3114	0.4264	0.1913	0.304	0.09744
7	13.28	20.28	87.32	545.2	0.1041	0.1436	0.09847	0.06158	0.1974	0.06782
8	20.59	21.24	137.8	1320.0	0.1085	0.1644	0.2188	0.1121	0.1848	0.06222
9	12.18	20.52	77.22	458.7	0.08013	0.04038	0.02383	0.0177	0.1739	0.05677
10	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697

B.3 Audit Trace: Fold 3 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	7.691	33.81	47.92	170.4	0.1447	0.3454	0.0	0.0	0.304	0.09744
2	7.691	33.81	186.9	170.4	0.1447	0.3454	0.0	0.0	0.304	0.09744
3	7.691	33.81	47.92	170.4	0.05263	0.01938	0.0	0.0	0.106	0.09744
4	7.691	33.81	186.9	170.4	0.1447	0.01938	0.0	0.0	0.304	0.09744
5	27.42	10.38	186.9	2501.0	0.1447	0.01938	0.4268	0.2012	0.106	0.04996
6	7.691	10.38	47.92	170.4	0.1447	0.3454	0.4268	0.0	0.304	0.09744
7	10.97	17.2	71.73	371.5	0.08915	0.1113	0.09457	0.03613	0.1489	0.0664
8	17.29	22.13	114.4	947.8	0.08999	0.1273	0.09697	0.07507	0.2108	0.05464
9	20.31	27.06	132.9	1288.0	0.1	0.1088	0.1519	0.09333	0.1814	0.05572
10	11.9	14.65	78.11	432.8	0.1152	0.1296	0.0371	0.03003	0.1995	0.07839

B.4 Audit Trace: Fold 4 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	10.38	43.79	143.5	0.1634	0.01938	0.0	0.0	0.1203	0.09744
2	6.981	10.38	43.79	143.5	0.05263	0.01938	0.0	0.2012	0.1203	0.04996
3	6.981	39.28	43.79	143.5	0.05263	0.01938	0.0	0.0	0.1203	0.04996
4	28.11	39.28	43.79	2501.0	0.05263	0.01938	0.4268	0.0	0.1203	0.04996
5	28.11	10.38	43.79	143.5	0.05263	0.01938	0.0	0.0	0.1203	0.04996
6	6.981	39.28	43.79	2501.0	0.05263	0.3454	0.4268	0.2012	0.304	0.09744
7	20.09	23.86	134.7	1247.0	0.108	0.1838	0.2283	0.128	0.2249	0.07469
8	11.52	18.75	73.34	409.0	0.09524	0.05473	0.03036	0.02278	0.192	0.05907
9	19.17	24.8	132.4	1123.0	0.0974	0.2458	0.2065	0.1118	0.2397	0.078
10	17.75	28.03	117.3	981.6	0.09997	0.1314	0.1698	0.08293	0.1713	0.05916

B.5 Audit Trace: Fold 5 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	9.71	43.79	143.5	0.1447	0.3454	0.0	0.0	0.2906	0.04996
2	6.981	9.71	43.79	143.5	0.1447	0.3454	0.0	0.0	0.2906	0.04996
3	6.981	39.28	43.79	143.5	0.05263	0.01938	0.0	0.0	0.2906	0.04996
4	6.981	9.71	43.79	143.5	0.05263	0.01938	0.4268	0.2012	0.2906	0.04996
5	6.981	9.71	43.79	143.5	0.1447	0.3454	0.0	0.0	0.2906	0.04996
6	28.11	9.71	188.5	2501.0	0.1447	0.3454	0.4268	0.2012	0.2906	0.09744
7	8.726	15.83	55.84	230.9	0.115	0.08201	0.04132	0.01924	0.1649	0.07633
8	19.4	18.18	127.2	1145.0	0.1037	0.1442	0.1626	0.09464	0.1893	0.05892
9	19.73	19.82	130.7	1206.0	0.1062	0.1849	0.2417	0.0974	0.1733	0.06697
10	12.19	13.29	79.08	455.8	0.1066	0.09509	0.02855	0.02882	0.188	0.06471

B.6 Audit Trace: Fold 6 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	7.691	33.81	47.92	170.4	0.05263	0.3454	0.0	0.0	0.2906	0.09744
2	28.11	9.71	47.92	2501.0	0.1447	0.02344	0.4268	0.2012	0.2906	0.09744
3	28.11	33.81	188.5	2501.0	0.05263	0.02344	0.0	0.0	0.106	0.04996
4	7.691	33.81	47.92	2501.0	0.05263	0.02344	0.0	0.0	0.106	0.09744
5	7.691	9.71	47.92	2501.0	0.05263	0.02344	0.0	0.0	0.106	0.09744
6	7.691	33.81	47.92	2501.0	0.1447	0.02344	0.4268	0.0	0.2906	0.09744
7	11.6	12.84	74.34	412.6	0.08983	0.07525	0.04196	0.0335	0.162	0.06582
8	13.0	21.82	87.5	519.8	0.1273	0.1932	0.1859	0.09353	0.235	0.07389
9	15.46	11.89	102.5	736.9	0.1257	0.1555	0.2032	0.1097	0.1966	0.07069
10	14.9	22.53	102.1	685.0	0.09947	0.2225	0.2733	0.09711	0.2041	0.06898

B.7 Audit Trace: Fold 7 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	39.28	43.79	143.5	0.05263	0.3114	0.0	0.0	0.106	0.09744
2	6.981	39.28	43.79	143.5	0.1634	0.01938	0.0	0.0	0.304	0.04996
3	28.11	9.71	43.79	143.5	0.05263	0.01938	0.0	0.0	0.106	0.04996
4	6.981	9.71	43.79	2501.0	0.1634	0.01938	0.0	0.0	0.106	0.09744
5	6.981	9.71	43.79	2501.0	0.1634	0.3114	0.0	0.0	0.304	0.09744
6	28.11	39.28	43.79	2501.0	0.05263	0.3114	0.4264	0.1913	0.304	0.09744
7	19.89	20.26	130.5	1214.0	0.1037	0.131	0.1411	0.09431	0.1802	0.06188
8	17.42	25.56	114.5	948.0	0.1006	0.1146	0.1682	0.06597	0.1308	0.05866
9	11.95	14.96	77.23	426.7	0.1158	0.1206	0.01171	0.01787	0.2459	0.06581
10	14.69	13.98	98.22	656.1	0.1031	0.1836	0.145	0.063	0.2086	0.07406

B.8 Audit Trace: Fold 8 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	9.71	43.79	2501.0	0.06251	0.01938	0.0	0.2012	0.304	0.09744
2	6.981	9.71	43.79	143.5	0.1634	0.01938	0.4268	0.0	0.304	0.04996
3	28.11	9.71	43.79	2501.0	0.06251	0.01938	0.0	0.0	0.304	0.04996
4	28.11	9.71	43.79	143.5	0.06251	0.3454	0.0	0.0	0.106	0.09744
5	28.11	39.28	188.5	2501.0	0.1634	0.01938	0.0	0.0	0.106	0.09744
6	6.981	9.71	43.79	143.5	0.1634	0.3454	0.4268	0.0	0.304	0.09744
7	13.05	18.59	85.09	512.0	0.1082	0.1304	0.09603	0.05603	0.2035	0.06501
8	10.16	19.59	64.73	311.7	0.1003	0.07504	0.005025	0.01116	0.1791	0.06331
9	16.24	18.77	108.8	805.1	0.1066	0.1802	0.1948	0.09052	0.1876	0.06684
10	14.87	20.21	96.12	680.9	0.09587	0.08345	0.06824	0.04951	0.1487	0.05748

B.9 Audit Trace: Fold 9 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	33.81	43.79	143.5	0.05263	0.3454	0.0	0.0	0.304	0.09744
2	6.981	9.71	43.79	143.5	0.05263	0.01938	0.0	0.2012	0.106	0.09744
3	28.11	33.81	43.79	2501.0	0.05263	0.01938	0.0	0.0	0.304	0.09744
4	6.981	9.71	43.79	2501.0	0.05263	0.01938	0.4268	0.2012	0.106	0.09744
5	6.981	9.71	43.79	2501.0	0.05263	0.01938	0.0	0.2012	0.106	0.05024
6	6.981	33.81	43.79	143.5	0.05263	0.3454	0.0	0.0	0.304	0.09744
7	14.97	16.95	96.22	685.9	0.09855	0.07885	0.02602	0.03781	0.178	0.0565
8	11.43	15.39	73.06	399.8	0.09639	0.06889	0.03503	0.02875	0.1734	0.05865
9	11.81	17.39	75.27	428.9	0.1007	0.05562	0.02353	0.01553	0.1718	0.0578
10	10.57	18.32	66.82	340.9	0.08142	0.04462	0.01993	0.01111	0.2372	0.05768

B.10 Audit Trace: Fold 10 | Gen: 100%

#	radius1	texture1	perimeter1	area1	smoothness1	compactness1	concavity1	concave_points1	symmetry1	fractal_dimension1
1	6.981	9.71	43.79	143.5	0.1634	0.3114	0.0	0.0	0.2743	0.09744
2	6.981	9.71	43.79	143.5	0.1634	0.3114	0.0	0.1913	0.2743	0.04996
3	28.11	39.28	43.79	2501.0	0.05263	0.0265	0.4264	0.0	0.1167	0.04996
4	6.981	9.71	43.79	143.5	0.1634	0.3114	0.0	0.0	0.2743	0.04996
5	6.981	9.71	43.79	2501.0	0.1634	0.0265	0.0	0.0	0.2743	0.04996
6	6.981	39.28	43.79	2501.0	0.1634	0.0265	0.4264	0.0	0.2743	0.09744
7	13.71	18.68	88.73	571.0	0.09916	0.107	0.05385	0.03783	0.1714	0.06843
8	11.94	18.24	75.71	437.6	0.08261	0.04751	0.01972	0.01349	0.1868	0.0611
9	10.51	23.09	66.85	334.2	0.1015	0.06797	0.02495	0.01875	0.1695	0.06556
10	11.75	17.56	75.89	422.9	0.1073	0.09713	0.05282	0.0444	0.1598	0.06677